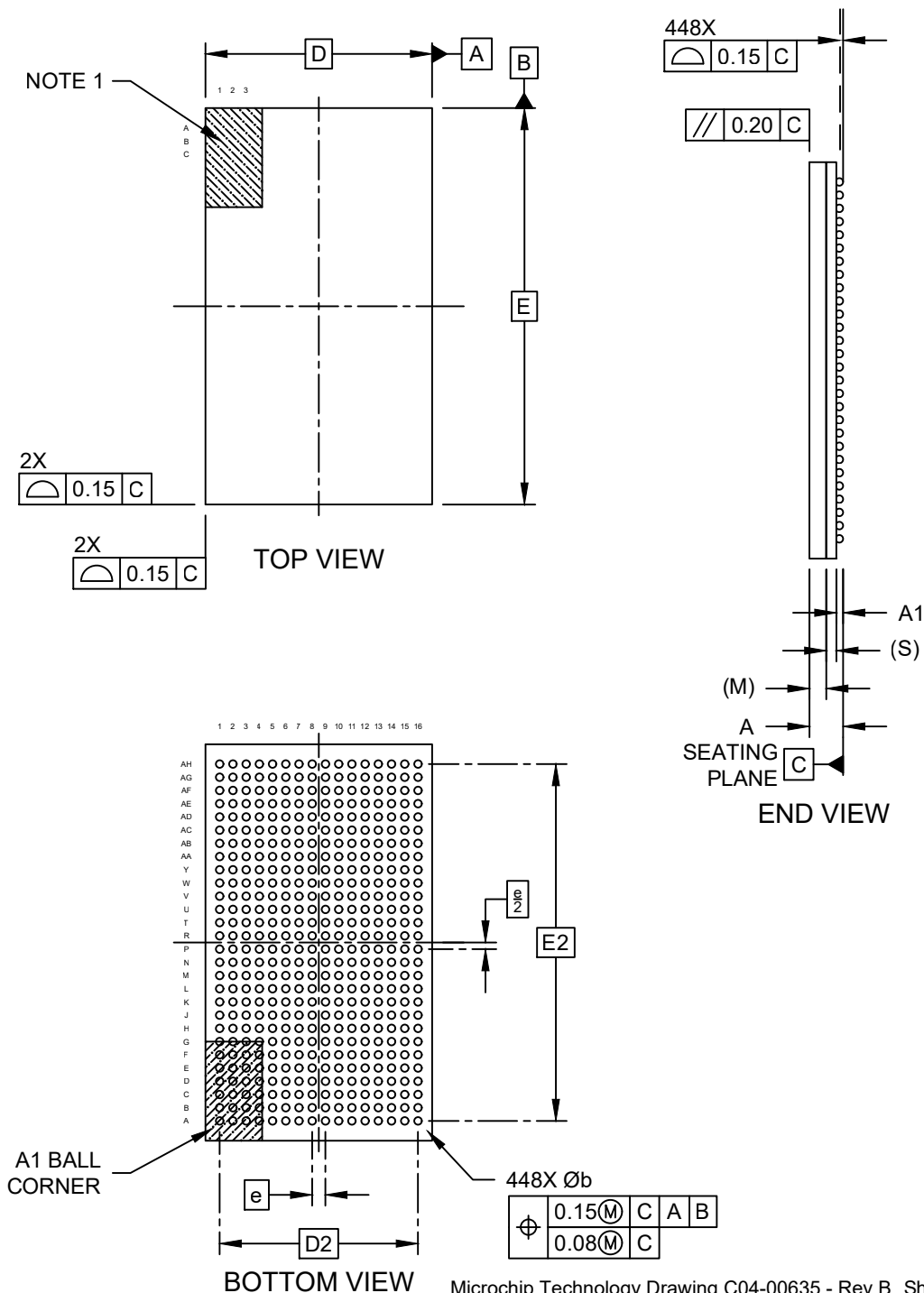


Packaging Diagrams and Parameters

448-Ball Fine-Pitch Ball Grid Array Package (6VW) - 12x21x1.78 mm Body [FBGA]

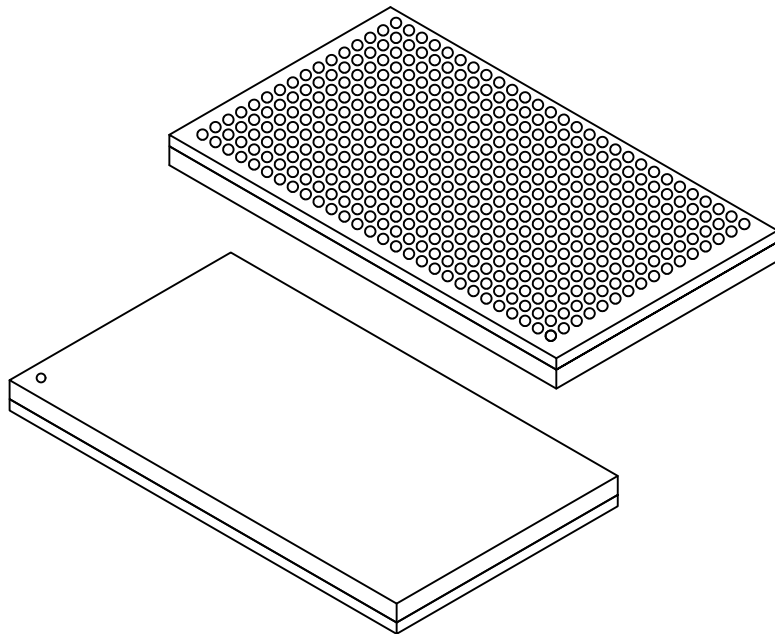
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

448-Ball Fine-Pitch Ball Grid Array Package (6VW) - 12x21x1.78 mm Body [FBGA]

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Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	448		
Pitch	e	0.70 BSC		
Overall Height	A	—	1.68	1.78
Ball Height	A1	0.25	0.30	0.35
Mold Thickness	M	0.80 REF		
Substrate Thickness	S	0.57 REF		
Overall Length	D	12.00 BSC		
Ball Array Length	D2	10.50 BSC		
Overall Width	E	21.00 BSC		
Ball Array Width	E2	18.90 BSC		
Ball Diameter	b	0.35	0.40	0.45

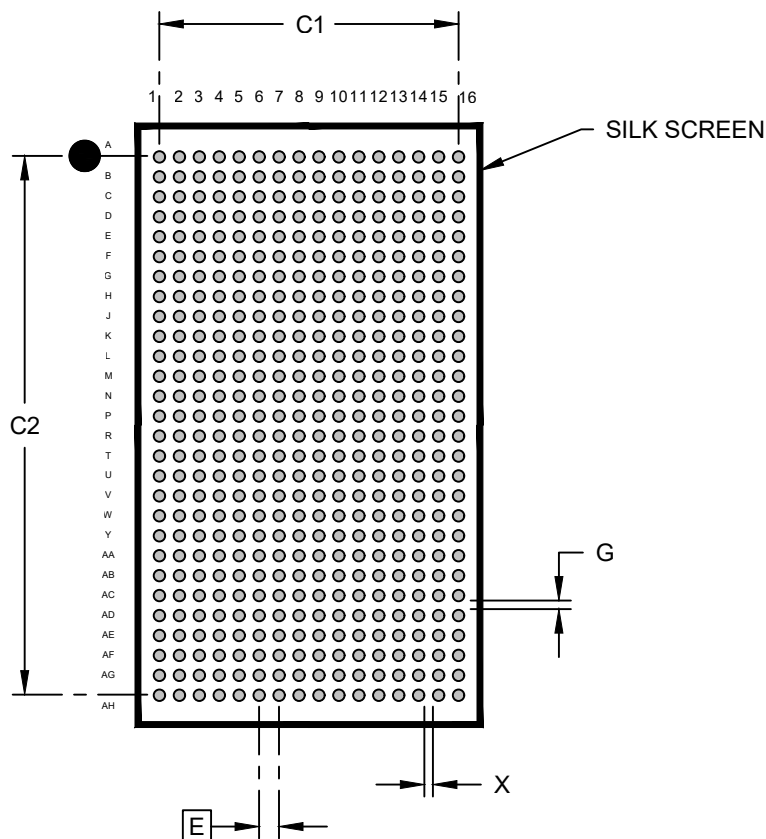
Notes:

- Pin 1 visual index feature may vary but must be located within the hatched area.
- Package is saw singulated.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

448-Ball Fine-Pitch Ball Grid Array Package (6VW) - 12x21x1.78 mm Body [FBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.70 BSC		
Contact Pad Spacing	C1		10.50	
Contact Pad Spacing	C2		18.90	
Contact Pad Diameter (X448)	X			0.35
Contact Pad to Contact Pad	G	0.30		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during the reflow process.